

Agent Trajectory

Task: What's the population of the capital of France?



u_0 User Query

What's the population of the capital of France?



y_1 Reasoning/Action

...I should search for the capital of France first



o_1 Tool/Observation

Query: capital of France
Result: Paris



y_1 Reasoning/Action

...the population of Paris is 2,060,000....



o_2 User Feedback

...I want to further know the population in 2024...

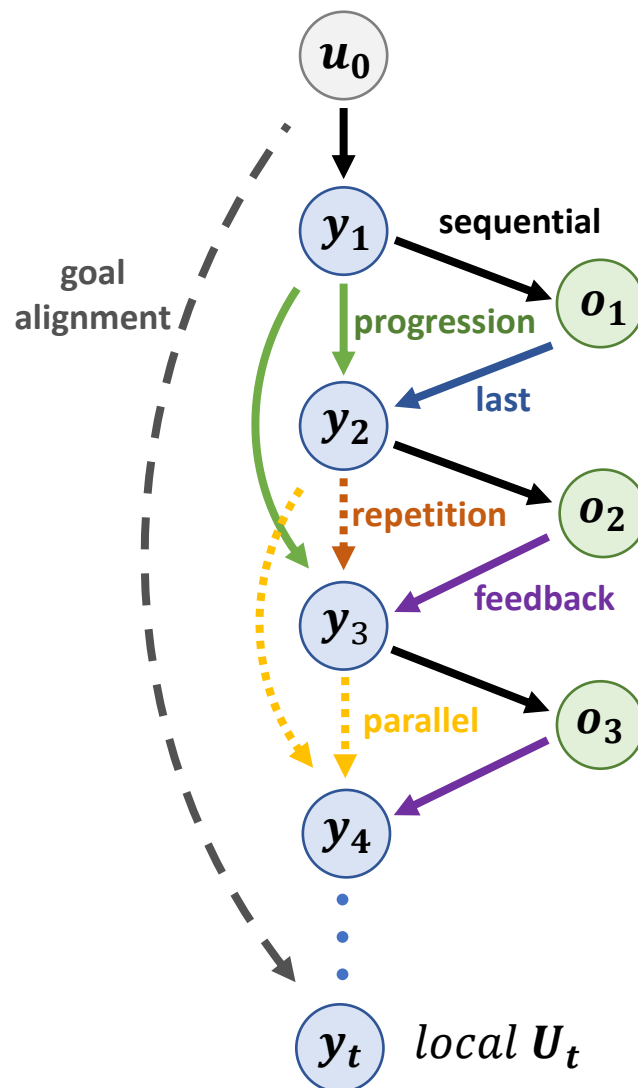
...



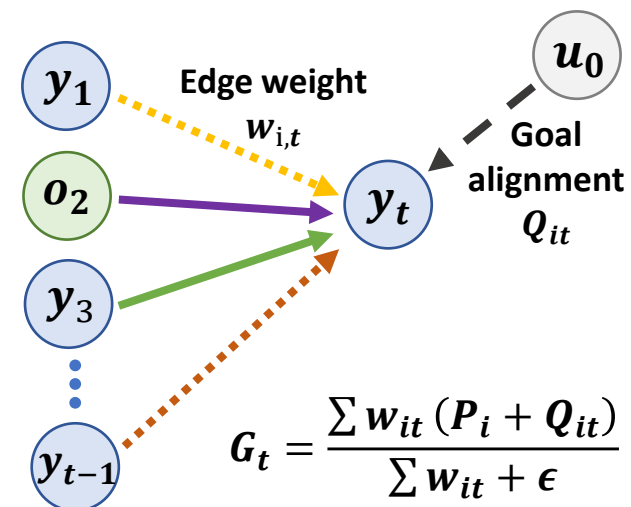
y_n Reasoning/Action

Population of the capital of France in 2024 is 2,161,000.

Graph Construction



Uncertainty Propagation



Propagation with momentum

$$H_t = \eta_g G_t + \eta_m m_t$$

↑
Historical risk

↑
momentum

Uncertainty Estimation

local U_t

History H_t

$$R_t = \lambda_u U_t + \lambda_h H_t$$